

Minor Connectors

Learning objectives:

- Identify the functions of minor connectors in RPD
 - Distinguish the different types of minor connectors in terms of shape and position.
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Definition

Minor connector: a unit of a partial denture that connects the major connector to other components (e.g., direct retainer, indirect retainer, denture base).

Functions

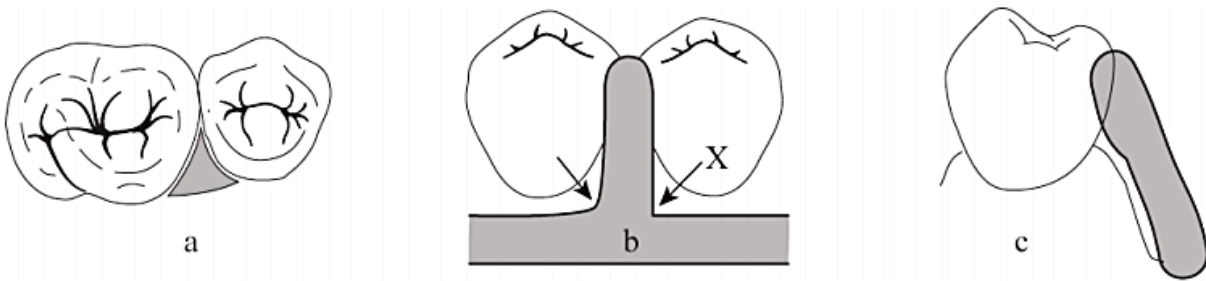
1. Provide unification and rigidity.
2. Provide stress distribution by transferring stresses from the major connector to other parts of the partial denture and from the partial denture to the abutment teeth.
3. Act as bracing elements by contacting the guiding planes opposing the retentive arms.
4. Maintain a path of insertion by contacting the guiding planes.

Basic Types of Minor Connectors:

1. **Embrasure minor connector** (between two adjacent teeth):

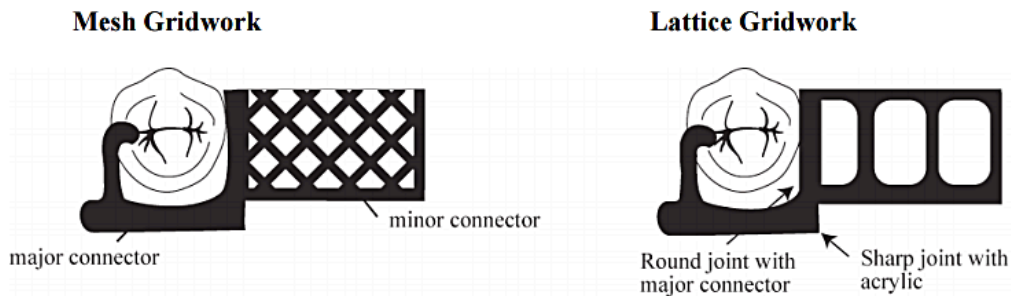
Shape and position

- (a) These connectors should be somewhat triangular-shaped in cross-section to minimise intrusion into the tongue or vestibular spaces, while still providing adequate bulk for rigidity. A minor connector should fill the embrasure space so that a smooth surface is presented to the tongue and so that areas where food can be trapped are minimised.



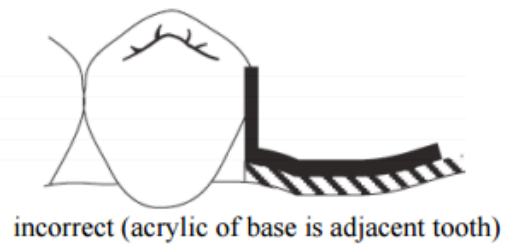
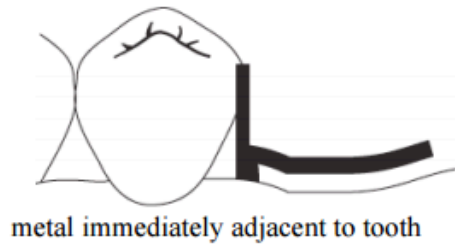
- (b) A minor connector should join the major connector at a right angle and cover as small an area of tissue as possible. The juncture to the major connector should be rounded (arrow) not sharp (X) unless the juncture includes an acrylic finish line.
- (c) Relief should be placed on the master cast so that the minor connector does not lie directly on the soft tissue.

2. Gridwork minor connectors: cover the edentulous area.

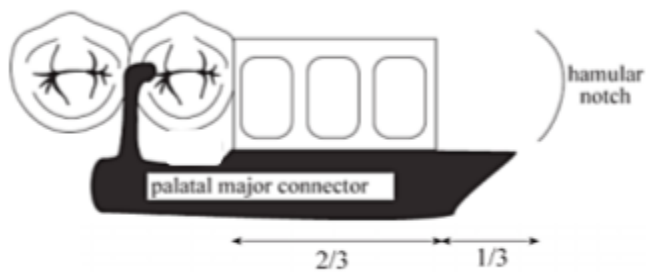


Shape and position:

- It should be rigid enough to support and resist breakage of the denture base.
- It is available in many forms, the most common are:
 1. Mesh gridwork construction
 2. Lattice gridwork construction.
- It should allow proper arrangement of natural teeth.
- Adequate mechanical retention of the denture base resin is gained by providing relief under the minor connector gridwork to allow the acrylic resin to flow under the gridwork. Relief under the gridwork should begin 1.5 - 2 mm from the abutment tooth. This will create a metal-to-tissue contact immediately adjacent to the tooth. A metal surface is preferable since it wears less, and is less porous, thus facilitating hygiene.



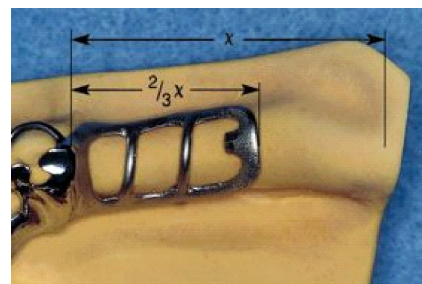
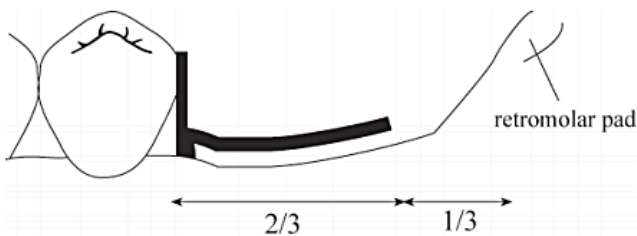
- Maxillary distal extension gridworks should extend at least $\frac{2}{3}$ of the length of the ridge to the hamular notch. However, the junction or finishing line of the *maxillary major connector* should extend fully to point to the hamular notch area.



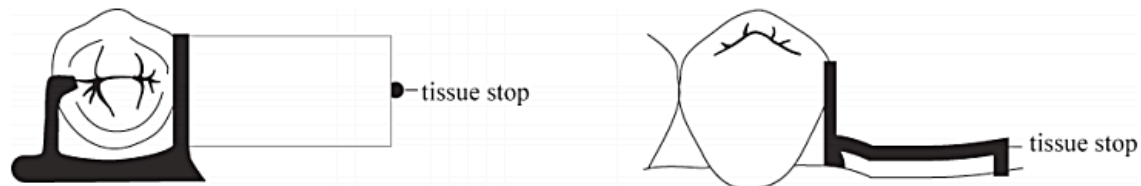
- Lingually, the junction of the maxillary major connector with the gridwork should be approximately 2 mm medial to an imaginary line along the lingual surfaces of the replacement denture teeth. This ensures a bulk of acrylic surrounding the replacement denture teeth to provide an adequate bond to the denture base.



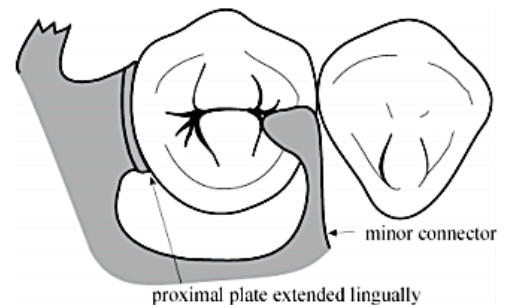
- Facially, the Gridwork minor connector ends just over the crest of the residual ridge.
- In mandibular distal extension cases, the minor connector should cover $\frac{2}{3}$ of the length of the edentulous ridge.



- Mandibular distal extension gridworks should have a “tissue stop” at their posterior limit. This is an extension of the metal through the relief wax providing direct contact with the ridge. A tissue stop, in theory, prevents framework deformation and its consequences. Maxillary distal extensions do not need tissue stops, as maxillary major connectors act as one.



- Minor connectors originating from the gridwork in an edentulous area usually take the form of vertical metal plates (proximal plates) that make broad contact with prepared guiding planes. These proximal plates may or may not terminate in an occlusal rest, depending on the partial denture design. The plate is shifted slightly towards the lingual to increase rigidity, enhance reciprocation and improve esthetics.



Finish lines

The junction of acrylic resin to the major connector is called a finish line; they must be less than 90 degrees with a slight undercut in the metal. This provides for maximum bulk of the acrylic resin denture base at the metal junction, to prevent the creation of thin, weak, feather edges which can easily fracture or distort.

